

Appendix 1

Methods of Powder Characterization and Testing

Standard Title	Origin				
	ISO	ASTM	MPIF	CIS	Other
Metallic powders—determination of apparent density—Part 1: Funnel method	3923/1	B 212	04	GOST 19440-94	...
Density, apparent of metal powders using the Arnold meter	...	B 703	48	...	...
Density, apparent of non-free-flowing metal powders	...	B 417	28	...	...
Metallic powder—determination of apparent density—Part 2: Scott volumeter method	3923/2	B 329	...	GOST 19440-94	...
Test method for metal powder skeletal density by helium or nitrogen pycnometry	...	B923	...	...	...
Flow rate of metal powders	4490	B 213	03	GOST 20899-75	...
Test methods for flow rate of metal powders using the Carney funnel	...	B964	...	...	...
Measuring dimensional change of metal powder specimens due to sintering	4492	B 610	44	GOST 29012-91	...
Metallic powders—determination of oxygen content by reduction methods—Part 1, part 2, part 3, part 4	4491-1, 2, 3, 4	E159	02	GOST 29006-91 GOST 27417-87 GOST 18897-73	...
Standard test method for loss of mass in a reducing gas atmosphere for cobalt, copper, tungsten, and iron powders (hydrogen loss)	4491/2	E159	02	GOST 29006-91	...
Test method for determination of hydrogen in aluminum and aluminum alloys by inert gas fusion	...	E2792	...	...	...
Test method for determination of hydrogen in titanium and titanium alloys by inert gas fusion thermal conductivity/infrared detection method	...	E1447	...	...	...
Test method for analysis of titanium and titanium alloys by direct current plasma and inductively coupled plasma atomic emission spectrometry	...	E2371	...	...	...
Test method for analysis of Ni-base alloys by wavelength dispersive X-ray fluorescence spectrometry	...	E2465	...	...	...
Test method for analysis of nickel alloys by inductively coupled plasma atomic emission spectrometry (performance-based method)	...	E2594	...	...	...

Standard Title	Origin				
	ISO	ASTM	MPIF	CIS	Other
Test method for determination of carbon in refractory and reactive metals and their alloys by combustion analysis	...	E1941	...	...	...
Metallic powders—determination of envelope—specific surface area from measurements of the permeability to air of a powder bed under study-state flow conditions	10070	...	...	GOST 23401-90	...
Metallic powders—determination of particle size distribution by gravitational sedimentation in a liquid and attenuation measurement	10076	...	...	...	...
Metallic powders—determination of acid insoluble content in iron, copper, tin, and bronze powders	4496	E194	6	GOST 30550-98	...
Terminology relating to sieves, sieving methods, and screening media	...	E1638	...	...	...
Metallic powders—determination of particle size by dry sieving	4497	B 214	05	GOST 18318-94	MIL-STD-1233
Particle size analysis—laser diffraction methods—Part 1: General principles	13320-1				
Metallic powders. determination of particle shape	...	...	...	GOST-25849-83	...
Test method for determination of oxygen in tantalum powder by inert gas fusion technique	...	E1569	...	...	...
Metallic powders, excluding powders for hardmetals—determination of compactibility (compressibility) in uniaxial compression	3927	B 331	45	GOST 25290-90	...
Metallic powders—determination of Green strength by transverse rupture of rectangular compacts	3995	B 312	15	GOST 25282-93	...
Particle size distribution of metal powders and related compounds by light scattering	...	B 822	...	...	...
Particle size distribution of refractory metals and their compounds by X-ray monitoring of gravity sedimentation	...	B 761	...	...	...
De-agglomeration of refractory metal powders and their compounds prior to particle size analysis	...	B 859	...	...	...
Particle size distribution of refractory metals-type powders by turbidimetry	...	B 430	...	...	...
Particle size, average, of metal powders by Fisher subsieve sizer	...	B 330 C 721	32	...	MIL-STD-1233
Particle size analysis of particulate substances by optical microscopy		E 20	...	GOST 23402-78	...
Standard test methods for determining average grain size	...	E112	...	...	...
Permeable sintered metal materials—determination of bubble test pore size	4003	...	...	GOST 26849-86	...
Pore volume distribution of catalysts by mercury intrusion porosimetry	...	D 4284	...	...	...
Bulk density and porosity of granular refractory materials by mercury intrusion porosimetry	...	C 493	...	...	...
Standard test method for apparent porosity in cemented carbides	...	B276	...	...	...

Standard Title	Origin				
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Sampling and sample preparation procedures	3082	...	...	GOST 23148 ...	
Sampling and testing aluminum powder and paste	...	D 480	...	...	...
Sampling finished lots of metal powders	3954	B 215	01	GOST 23148-78	
Standard shear testing method for bulk solids using the Jenike shear cel	...	D 6128	...	...	...
Subsieve analysis of granular metal powders by air classification	...	B 293	12	...	...
Tap density of metal powders	3953	B 527	46	GOST 25279-93	MIL-STD-1233
Tension test specimens for pressed and sintered metal powders	2740	E 8	10	...	...
Testing copper-base infiltrating powders	...	...	49	...	...
Terms used in powder metallurgy	3252	B 243	9	GOST 17359-82	
British Standards 2955 glossary of terms relating to powders	...	...	...	...	British Standards 2955
Wire-cloth sieves for testing purposes, specification	565 3310/1	E 11		GOST 3584	RR-S-366
Standards referenced include the following: American Society for Testing and Materials (ASTM); International Organization for Standardization (ISO); Metal Powder Industries Federation (MPIF); Commonwealth of Independent States (CIS); MIL—Superintendent of Documents, US Government Printing Office, Washington, DC 20402.					